EXPERIMENT 02

Finite State Automaton (DFA) for Strings Ending with 'ab'

AIM

To design and implement a Deterministic Finite Automaton (DFA) in Python that accepts the language L = {w ∈ {a, b}* | w ends with 'ab'}.

PROCEDURE

1. Define the alphabet set Σ = {'a', 'b'} and states Q = {q0, q1, q2}, where q0 is the start state and q2 is the accepting state.
2. Construct the transition function δ: Q × Σ → Q: δ(q0, 'a') = q1, δ(q0, 'b') = q0; δ(q1, 'a') = q1, δ(q1, 'b') = q2; δ(q2, 'a') = q1, δ(q2, 'b') = q0.
3. Implement a Python class or function that iteratively transitions through states for each character in the input string.
4. Verify whether the final state reached is the accepting state q2.
5. Test the automaton on valid and invalid strings to validate recognition.

PROGRAM

```
class DFAEndsWithAB:
    def __init__(self):
        self.start_state = 'q0'
        self.accept_states = {'q2'}
        self.transitions = {
            ('q0', 'a'): 'q1',
            ('q0', 'b'): 'q0',
            ('q1', 'a'): 'q1',
            ('q1', 'b'): 'q2',
            ('q2', 'a'): 'q1',
            ('q2', 'b'): 'q0'
        }

    def process(self, input_str):
        current = self.start_state
        trace = [current]
        for char in input_str:
            if (current, char) not in self.transitions:
                return False, f"Invalid symbol '{char}' in alphabet"
            current = self.transitions[(current, char)]
            trace.append(current)

        is_accepted = current in self.accept_states
        return is_accepted, trace

dfa = DFAEndsWithAB()
```

```
test_strings = ["ab", "aab", "bab", "abba", "abbab", "bba", "b"]

print("--- DFA Execution Results ---")
for s in test_strings:
    accepted, trace = dfa.process(s)
    status = "ACCEPTED" if accepted else "REJECTED"
    print(f"String: {s:<8} | State Path: {' -> '.join(trace):<20} | Result:
{status}")
```

OUTPUT

```
> python3 experiment_02.py
--- DFA Execution Results ---
String: ab       | State Path: q0 -> q1 -> q2         | Result: ACCEPTED
String: aab      | State Path: q0 -> q1 -> q1 -> q2   | Result: ACCEPTED
String: bab      | State Path: q0 -> q0 -> q1 -> q2   | Result: ACCEPTED
String: abba     | State Path: q0 -> q1 -> q2 -> q0 -> q1 | Result: REJECTED
String: abbab    | State Path: q0 -> q1 -> q2 -> q0 -> q1 -> q2 | Result: ACCEPTED
String: bba      | State Path: q0 -> q0 -> q0 -> q1   | Result: REJECTED
String: b        | State Path: q0 -> q0               | Result: REJECTED
[Process completed with exit code 0]
```

RESULT

Thus, the finite state automaton recognizing strings ending with 'ab' was successfully designed, implemented in Python, and verified against diverse test cases.